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RESEARCH PAPER

Large Language Models for Process Knowledge Acquisition

Designing and Evaluating a Multi-agent System

Malik Schinckus · Anthony Simonofski · Nicolás Bono Rosselló

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Abstract Acquiring process knowledge remains a central challenge in business process management, particularly when process discovery approaches rely on manual elicitation and analysis. Following design science research, this paper proposes a theoretically grounded and empirically validated approach to support process knowledge acquisition using Large Language Models (LLMs). The paper outlines three main contributions. First, drawing from knowledge acquisition theory, 19 design requirements are defined for using LLMs in the knowledge acquisition process. Second, these are instantiated in a proposal, named PKAI, which is a novel multi-agent system that operationalizes the stages of preparation, socialization, and externalization in process discovery through specialized LLM-based agents. Third, the study provides empirical evidence of the benefits and limitations of the approach through three evaluation rounds: (i) a demonstration involving business process analysts validating the design requirements and their mapping to the instantiation, (ii) a quasi-experimental study highlighting that process analysts supported by PKAI perform better in designing conceptual models in semantic and pragmatic dimensions, and (iii) a real-world illustrative case study demonstrating the approach's applicability under business complexity and its impact on the knowledge acquisition process. The paper provides the first LLM-based artifact instantiation spanning the whole knowledge acquisition process.

Keywords Business process discovery · Knowledge acquisition · Conceptual modeling · Large language models · Business process management · Design science research

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M. Schinckus (✉) · A. Simonofski · N. Bono Rosselló
Namur Digital Institute, University of Namur, Rue de Bruxelles 61, 5000 Namur, Belgium
e-mail: malik.schinckus@unamur.be

A. Simonofski
e-mail: anthony.simonofski@unamur.be

N. Bono Rosselló
e-mail: nicolas.bonorossello@unamur.be

1 Introduction

Business process management promises several benefits such as performance management (Prilla and Nolte 2012; Buh et al. 2015), organizational flexibility (Badakhshan et al. 2020), strategic alignment (Sousa et al. 2011), or operational efficiency (Fehrer et al. 2022). Formalized process knowledge is fundamental for managing business processes and achieving those benefits (Malinova and Mendling 2018). The literature mentions three approaches to acquiring and formalizing business process knowledge: *research*, *automated discovery*, and *elicitation* (Antunes et al. 2020). *Research* involves manual analysis of business data without direct interaction with the most relevant stakeholders. This includes the analysis of various forms of business documentation and communication, such as specifications or spreadsheets. *Automated discovery* relies on automated data retrieval systems enhanced by different analysis techniques (van der Aalst 2010, 2012). For instance, process mining captures data traces of processes in information systems (Brock et al. 2024). *Elicitation* engages business process analysts and domain experts to identify and clarify key aspects of business processes. This involves techniques such as interviews, focus groups, or workshops (Dumas et al. 2013; Antunes et al. 2020).

While organizations express interest in *automated discovery*, widespread adoption of the approach remains limited among these approaches (Brock et al. 2024). Several challenges are observed such as system rigidity, poor tacit knowledge capture, and difficulties in managing granularity (van der Aalst et al. 2016; Wang et al. 2018; Beverungen et al. 2021). The need to capture contextual and tacit knowledge is widely acknowledged to bridge the gap between the conceptual model and reality (Beerepoot et al. 2023). This need could lead organizations to adopt more manual approaches, such as *elicitation* and *research*, to incorporate the unstructured knowledge of domain experts and thereby enhance process analysis (Simões et al. 2018). *Elicitation* is an essential backbone to business process discovery, initially to gain knowledge and later to validate the pre-formalized process knowledge. In addition, to prepare elicitation sessions or challenge their outcomes, business process analysts can leverage *research* to gain information and support conceptual modeling. When sufficient and accessible documentation is available, the value of this *research* approach is also particularly significant. Accordingly, the two approaches are utilized in a complementary manner and are both essential components of manually driven process discovery.

However, these two approaches suffer numerous challenges. *Elicitation* between process analysts and domain experts is ineffective; elicitation techniques are time-consuming and labor-intensive, requiring significant involvement of process analysts. Discussion can produce "copious and unwieldy data" (Cooke 1994) and then need several rounds of validation to abstract knowledge. Dutra et al. (2023) identified three main challenges: fragmented process knowledge, thinking of processes on a case level, and limited familiarity with business process modeling languages. In addition, inaccuracies occur when eliciting information subject to interpretation by analysts (Harman et al. 2016). In comparison, *research* is complex, sometimes requires specialized expertise, and is time-consuming (Herbst and Karagiannis 1999; Maqbool et al. 2019). Moreover, as the interpretation of knowledge acquired must be challenged, research is considered to stimulate later elicitation (Cooke 1994). Due to these challenges, research and elicitation require additional support. Better resource allocation to these tasks would foster more sustainable and flexible process discovery practices, thereby strengthening the overall business process management approach.

In theory, beyond elicitation, knowledge acquisition encompasses the formalization of knowledge during the creation and validation of a conceptual process model (Cooke 1994), which serves as one output of business process discovery phases. Grounded by this founding theory, we distinguish process discovery into two distinct stages aligned with knowledge creation theory (Nonaka et al. 1994): socialization (1), which involves generating tacit knowledge from unstructured (tacit) knowledge, and externalization (2), which refers to formalizing knowledge, such as abstracting a process model. These two steps, often explored independently in literature, are closely interconnected, with the quality of formalized knowledge highly dependent on that of elicited knowledge. In the context of our study, the externalization phase is guided by the pragmatic objectives of knowledge formalization as expressed by practitioners. This provides a more objective foundation for analyzing the potential integration of emerging technologies, such as large language models.

Building on this research opportunity and recontextualizing business process discovery within the knowledge acquisition theory, this paper explores how to enhance the manually driven approaches (i.e., *research* and *elicitation*) by using generative Artificial Intelligence (AI) to support business process discovery. Large language models (LLMs) have the potential to transform process management by enabling process discovery. For example, they can analyze documentation, communication logs, and interactive queries (Vidgof et al. 2023). This paper aims to enhance the knowledge acquisition process through LLMs' capabilities. More concretely, we examine through a design science research approach how an LLM-based multi-agent system that relies on state-of-the-art LLM-enhancing techniques can support and impact process discovery by replicating the knowledge acquisition process. This broad objective is divided into three specific research questions:

(RQ1) What are the process knowledge acquisition challenges that LLMs can support? The objective is to develop a thorough understanding of current practices to identify where LLMs can assist practitioners in the knowledge acquisition process. To this end, we first formalize practitioners' current practices and map these to knowledge acquisition theory. We define 19 design requirements applicable to process analysts to ensure our proposal's relevance in supporting process discovery. The problem diagnosis and the formalization of these requirements, rooted in the context of business process management and grounded in knowledge acquisition theory, serves as a knowledge base to build artifacts aiming to support the knowledge acquisition process.

(RQ2) How to design LLMs-supported process knowledge acquisition? Research has not yet investigated the design choices involved in integrating LLMs to support knowledge acquisition. This research question addresses the current knowledge gaps in designing LLM-supported knowledge acquisition process through the development of a fully

functional artifact. We present and justify our implementation choices by demonstrating the fulfillment of our design requirements. We leverage LLMs to perform specific identified tasks within the knowledge acquisition process. Specifically, we operationalize the knowledge acquisition process through a solution that orchestrates its various formalized stages via a flexible multi-agent system, designed to support process analysts in business process discovery.

(RQ3) How does introducing LLMs impact the knowledge acquisition process? To date, there is no empirical evidence regarding the ability of LLMs to address the identified challenges. Therefore, we investigate the impact and relevance of leveraging LLMs to address these challenges. First, through empirical studies, we assess the ability of our proposal to collect process information from domain experts and to link natural language with business process modeling notation (BPMN). Based on the 3QM framework from Overhage et al. (2012), we empirically evaluate the process model quality, highlighting the semantic dimension. For instance, we demonstrate that our proposal enables the conceptualization of more relevant models while delegating the formalization of knowledge to domain experts enhanced by our proposed instantiation, named *PKAI*. We observe an improvement in the semantic and pragmatic quality of the models refined by process analysts, leveraging the knowledge elicited and pre-formalized by *PKAI*. To date, there has been a lack of extensive empirical evaluations regarding the introduction of LLMs in business process discovery; our goal is to provide this, grounded in the context of formalized current practices. Then, we investigate the impacts of introducing our supported knowledge acquisition proposal from a methodological point of view, and we illustrate how such a solution could be integrated in current practices. The most important demonstrated impact is the ability to support autonomous exploration of process knowledge by empowering expert domains up to an intermediate level of conceptualized and formalized knowledge. Process analysts and domain experts can work asynchronously, and the knowledge acquisition process is standardized, overcoming many of the drawbacks of manually driven approaches.

2 Theoretical Background and Related Work

2.1 Knowledge Acquisition Theory

Knowledge represents a crucial asset in organizations, often called intellectual capital, facilitating the development of core competencies and strategic insights (Alavi and Leidner 2001). Knowledge elicitation, part of the process of knowledge acquisition, is defined as "*the process of collecting from a human source of knowledge, information that is thought to be relevant to that knowledge*" (Cooke 1994, p. 803). Most of this knowledge resides with individuals rather than with the organization itself, making eliciting personal knowledge essential (Gavrilova and Andreeva 2012). Nonaka et al. (1994) argue that, while individuals develop new knowledge, organizations play a critical role in articulating and amplifying it.

In her founding theoretical knowledge acquisition paper, Cooke (1994, p. 802) considers that knowledge acquisition spans from elicitation to "*the explication and formalization of that knowledge*". However, this process presents challenges due to the difficulty of externalizing tacit knowledge, which is deeply embedded in domain experts, their practices, and experiences (Simões et al. 2016). Nonaka et al. (1994) state that converting from tacit knowledge to other tacit knowledge, for instance, sharing experience through a discussion, is called "*socialization*". The conversion of tacit knowledge into explicit knowledge (i.e., conceptual models) is called "*externalization*" (Nonaka et al. 1994). Therefore, this paper considers business process discovery as an application of the knowledge acquisition process in business process management.

Some proposals in the literature focus on facilitating the knowledge acquisition process. Elicitation between process analysts and domain experts often leads to information loss and inconsistencies in understanding domain-specific processes (Fleischmann et al. 2012). In response, Fleischmann et al. (2013) propose a subject-oriented business process management approach, enabling domain experts to describe workflows from their perspectives. Harman et al. (2015) suggest role-playing in virtual environments to simulate elicitation and contextualize processes. Kathleen et al. (2014) introduce a 3D storyboard solution for visualizing processes interactively, while Antunes et al. (2013), Brown et al. (2014), and Thuan et al. (2022) advocate storytelling-based methods using "scenes" and "stories" for accessible, narrative driven models. Simões et al. (2016, 2018) focus on eliciting tacit knowledge through personal and contextual insights, complementing formal models (Nonaka 2009). Finally, Hoppenbrouwers et al. (2018) propose collaborative workshops enabling domain experts to model processes.

Table 1 LLM-enhanced knowledge acquisition processes, limitations identified in related works

	Socialization	Externalization
Domain experts' requirements	Lack of proposals on LLM-based guided process knowledge socialization	Lack of guidance to allow process knowledge to meet analytical needs
Process analysts' requirements	Lack of empirical studies to assess the quality and validity of LLM-based elicited information	Lack of empirical assessments of the quality and usability of generated models

These proposals emphasize getting people to express ideas they do not usually discuss, thereby uncovering tacit knowledge often overlooked in traditional elicitation methods (Hill et al. 2015). While their focus is on the socialization aspects from the domain experts' point of view, they do not adequately assess the externalization of the outputs (e.g., conceptual models) generated during elicitation, nor do they guide the domain experts' discussion. Consequently, the elicited information is often limited to the most common process flows ("happy flow") or may lack relevant technical information, potentially rendering them insufficient for knowledge externalization from the analyst's point of view. These proposals provide limited support for externalization. Focused on domain expert requirements, they do not integrate design requirements from process analysts. Conversely, studies focused on externalization fail to incorporate the essential aspects of socialization, which hinders the establishment of precise requirements and results in evaluations of these proposals that are irrelevant and misaligned with the real context of use.

2.2 LLM-Supported Process Knowledge Acquisition

Trained on extensive volumes of unstructured data, LLMs have transformed natural language processing tasks, including text generation (Li et al. 2021) and code synthesis (Vidan and Fiedler 2023). They present significant opportunities for process discovery by analyzing documentation and facilitating interactive queries (Vidgof et al. 2023). However, their application in business process management is nascent, facing global challenges such as a high error rate in reflexive tasks (e.g., Nezhurina et al. 2024) or the phenomenon of hallucinations (Teubner et al. 2023). Despite ongoing research, clear guidelines for supporting conceptual modeling with LLMs remain underdeveloped (Fill et al. 2023). Additionally, the unpredictable behavior of LLMs underscores the need for standardizing output quality to enhance reliability in process modeling tools (Klievtsova et al. 2023). Zhou et al. (2023) highlight the variability in LLMs' behavior depending on prompt design. A key research focus is identifying requirements for LLMs to generate valuable outputs consistently. Focusing on structural consistency, some generative AI providers, such as OpenAI, Gemini, or Claude,[1] offer native functionalities that enforce structured outputs using formats like JSON. To address this volatility and improve the relevance, accuracy, and semantical consistency of LLMs' outputs, techniques such as prompt engineering (Wang et al. 2024) or retrieval-augmented generation (Chen et al. 2024) have been developed (Li et al. 2024). These techniques offer promising opportunities to integrate LLMs into specific tasks and address inherent limitations, yet no established guidelines or empirical evaluations currently exist for their use in business process discovery.

2.2.1 *LLM-Supported Knowledge Socialization*

Conversational agents for process modeling have shown promising results in enhancing knowledge elicitation. Klievtsova et al. (2023) and Vakayil et al. (2024) demonstrate how LLM-based chatbots can guide users through adaptive and interactive interviews, prompting them to provide detailed insights into business processes. This facilitates the collection of explicit knowledge and helps uncover tacit knowledge by interpreting user inputs, asking follow-up questions, and clarifying ambiguities. Similarly, Hu et al. (2024) explore how automated follow-up question generation, driven by LLMs, can improve the relevance and informativeness of responses during interviews, offering valuable design insights for future conversational agents. Jiang et al. (2023) highlight the potential of multi-agent chatbot systems in improving engagement and response quality, with LLMs enabling effective conversation management across various domains. More specialized interview techniques, such as laddering, are explored. Rietz and Maedche (2023) present *Ladderbot*, a text-based conversational agent capable of facilitating human-like laddering interviews. In this approach, LLMs help generate more detailed and structured responses, leading to deeper insights than traditional survey-based approaches. Overall, LLM-supported knowledge elicitation is a widely-discussed topic in recent years in different domains, such as requirements engineering (Ronanki et al. 2023). However, the elicitation of process-specific knowledge remains an unexplored area in the literature. Therefore, these

[1] Information retrieved from providers' public documentation, such as: https://platform.openai.com/docs/guides/structured-outputs?api-mode=responses, https://ai.google.dev/gemini-api/docs/structured-output, or https://docs.anthropic.com/en/docs/build-with-claude/prompt-engineering/prefill-claudes-response#example-structured-data-extraction-with-prefilling.

promising applications represent avenues for enhancing process knowledge elicitation, but further research and empirical evaluation are needed to actualize the impacts and capabilities of these agents in the business process management context.

2.2.2 *LLM-Supported Knowledge Externalization*

In recent years, LLMs have been explored for process knowledge externalization. Fill et al. (2023) and Cámara et al. (2023) conducted initial experiments on generating conceptual models using ChatGPT with UML or BPMN modeling languages. The preliminary findings reveal promising results but also highlight the LLMs' lack of deep understanding and pure intelligence needed to grasp the semantic complexities of the proposed models. Cámara et al. (2023) also highlight various syntactic deficiencies, response inconsistencies, and scalability challenges.

Mainly focusing on automated process modeling using BPMN language, the proposal from Kourani et al. (2024) automates process model generation from textual descriptions, emphasizing the role of prompt engineering techniques. Relying on practitioners' requirements, Ziche and Apruzzese (2024) develop a tailored chatbot to support process modelers in creating BPMN models. The proposal assists process modelers by guiding the resulting models' purpose, usage, and value creation, and utilizing embedded organizational documentation to tailor outputs specific to the organization's context. Then, Köpke and Safan (2024) present an efficient LLM-based modeling approach to reduce token consumption. Recent studies further indicate that models generated through the use of LLMs are often preferred to human-created models (Klievtsova et al. 2025).

Ongoing research delineates future research opportunities with proposals that are framed as automated solutions that replace or assist process analysts' work. They demonstrate promising results in externalizing textual knowledge already elicited with well-structured text. Nevertheless, despite promise, first insights show that the conceptual BPMN models produced by these proposals may not achieve the same level of quality as those produced by experienced process analysts. Therefore, it is necessary to recontextualize these promising approaches by aligning them with real-world practices and requirements. There is a need to assess how such a proposal can support knowledge acquisition processes. In addition to design requirements, there is a need to empirically study these state-of-the-art modeling capabilities, objectifying these capabilities, and comparing them to the current non-supported approach. Table 1 summarizes various limitations mentioned throughout Sect. 2.2.

3 Design Science Research Methodology

To address our research questions, we relied on a design science research (DSR) methodology (Hevner et al. 2004; Peffers et al. 2007). DSR allows us to improve our artifact proposal by following scientific and practitioner requirements iteratively, ensuring relevance and rigor for the proposed solution. According to Venable's foundational paper (2006), DSR enables iterative theory building across three key areas: problem diagnosis, technology invention or design, and technology evaluation. We iteratively conducted various data collections and analyses to refine problem-oriented design requirements and align our solution proposal (Fig. 1).

Drawing on the DSR from Peffers et al. (2007) (Fig. 2), we started with problem analysis. We identified statements and gaps in the *literature* on various current process discovery approaches (Sect. 1). We reviewed existing work supporting process discovery, focusing on knowledge elicitation or formalization. This step also involved an extensive study of existing solutions and their limitations (Sect. 2). Then, we formulated the problem statement. This included improving our understanding of the problem from *literature* and practitioners' insights to ensure the relevance of the design requirements. We challenged our problem statement and associated design requirements, relying on several data collections applied in the relevant environments explained in the research settings (Sect. 4.1). Through *in-depth interviews* with 11 experienced process analysts, we validated design requirements that guided our instantiation (Sect. 4). We then focused on the instantiation of our proposed solution (*PKAI*), drawing on the existing literature's artifacts and proposals (*literature review*) (Sect. 5). Then, we demonstrated that the solution implemented functionalities through *follow-up interviews* with a sub-sample of the process analysts (9 of the 11 interviewees) to validate the mapping of our design requirements to our instantiation. After that, to evaluate *PKAI* capabilities, we first performed a *quasi-experimental study* (n = 26) to compare knowledge acquisition process results with and without the support of *PKAI*. Finally, we directed *an illustrative case study* in a real organizational context with one process analyst and three domain experts to show the impacts of our proposal on current practices integrating a company's business complexity (Sect. 6). Before concluding, we positioned our contribution and solution proposal with relevant related works and detailed the limitations of our research (Sect. 7).

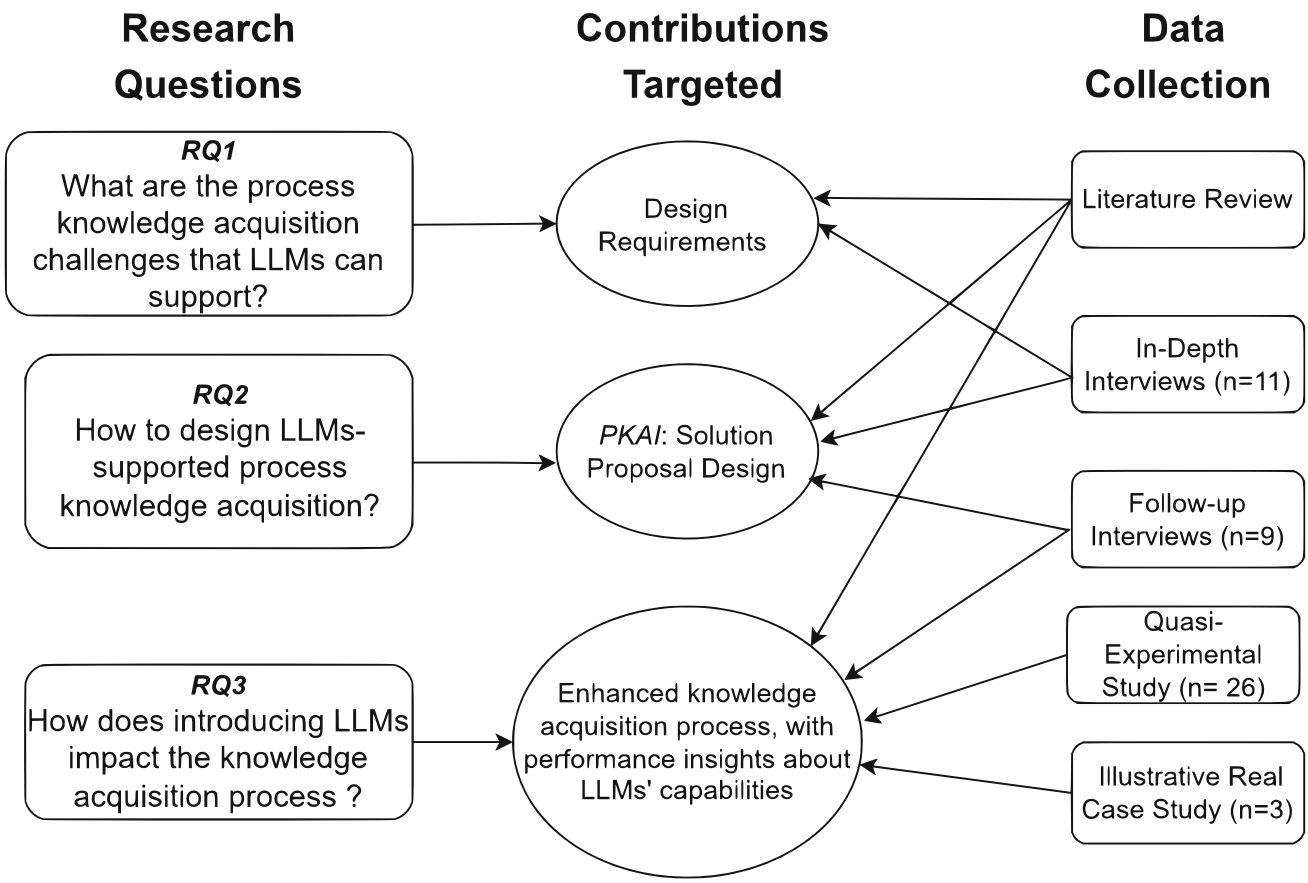


Fig. 1 Mapping between research questions (RQs), targeted contributions, and data collection techniques

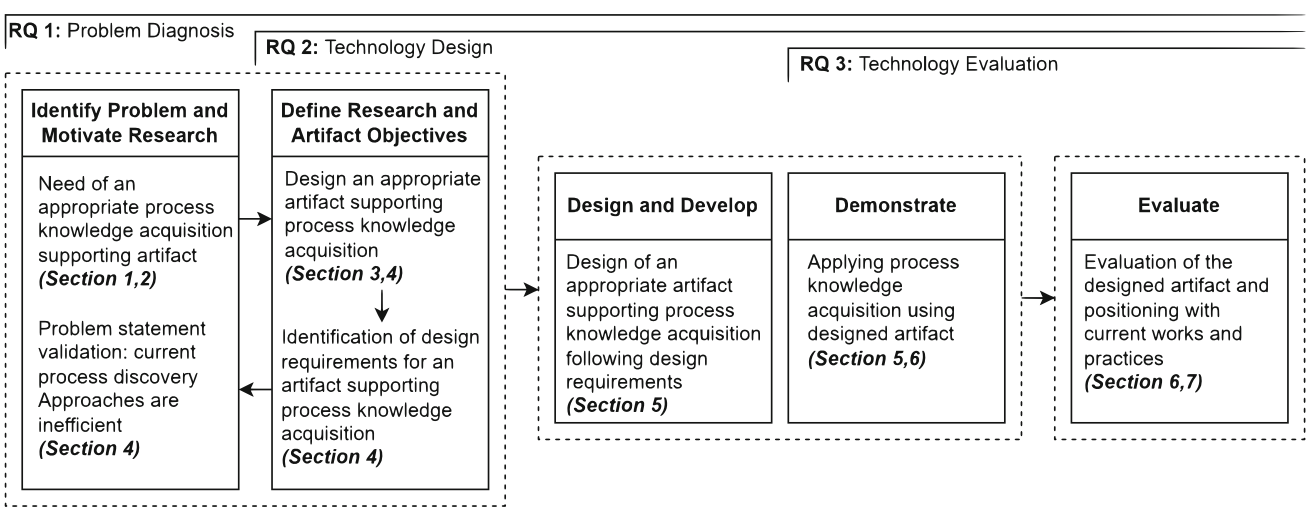


Fig. 2 Applied design science research methodology – adapted from Peffers et al. (2007)

4 Problem Statement and Definition of Requirements

4.1 Research Setting and Data Collection

To obtain grounded insights about the current knowledge acquisition process, one researcher joined the process steering committee of a process-based telecom company for 8 months. The researcher attended monthly "process forum" workshops following participant observation guidelines (van Maanen 2011). These two-hour meetings addressed business process management improvement opportunities. Then, the involved researcher, together with

Table 2 Information about the process analysts interviewed

ID	Process experience (years)	Current company sectors	Job title
#1	4	Medical	Business Process Analyst
#2	10	Legal	Business Process Architect
#3	9	IT & Digital Transformation	Program Manager
#4	7	Biotech	Business/ Functional Analyst
#5	14	Logistics	Business Analyst
#6	9	Telecoms	Business Process Analyst
#7	9	Logistics	Project Manager/Process Expert
#8	11	Mobility	Business Analyst
#9	9	IT & Digital Transformation	Business Process Analyst
#10	11	Medical	Business Process Architect
#11	8	Telecoms	Business Process Analyst

a fellow researcher, conducted a process management maturity audit, engaging with various stakeholders. This approach provided deeper insights into the barriers hindering the development of business process management practices, particularly in the business process discovery phase and the mechanisms of knowledge transfer between teams and departments. Building on the rigor developed through our research on the use of LLMs in business process management and the theoretical background of knowledge acquisition and creation theory, this immersion enabled us to understand the business process management development needs of an organization, ensuring relevance.

Considering this industry experience as prior contextual knowledge and to delve deeper into the knowledge acquisition process, we conducted 11 interviews (approx. 55–65 min). The interviewees are business process analysts with between 4 and 14 years of business process management experience. They all experienced process knowledge acquisition in different companies of different sizes and sectors in Belgium (see Table 2). These interviews were conducted in the form of meetings, following a semi-structured in-depth interview framework guided by design guidelines from Boyce and Neale (2006) and Guion et al. (2011). The research guide includes questions about process discovery practices and difficulties faced during the socialization and externalization phases.[2] As the qualitative research literature does not provide guidelines on how many participants to interview, our data collection was based on data saturation criteria (Guest et al. 2006).

We transcribed the interviews using the Whisper model from OpenAI. The transcriptions were then carefully reviewed and anonymized. Two researchers independently analyzed these interviews following the thematic content analysis method described in Mayring (2004) and Anderson (2007), using an inductive approach to define design requirements. The list of codes (meta-requirements) was iteratively extended when a new requirement was stated in the interviews. First, the analysis started by summarizing the interviews and think-aloud transcripts before recording them manually in a data memo. To code the data, we skimmed the transcripts and highlighted relevant sentences based on the research objective using "initial coding" guidelines from Saldana (2009). The created codes referred to requirements sub-categories identified in the interviews and presented below as the Challenges (*C*) to face. For each code, we linked a specific excerpt from the interview. Second, to give coherence to the coding of the first step, we performed axial coding (Corbin and Strauss 1990).

4.2 Problem Statement and Design Requirements

Some organizations rely on manually driven discovery approaches (i.e., *research* and *elicitation*). These approaches meet two objectives of the knowledge acquisition process. First, in an exploratory phase, process analysts require access to unstructured and sometimes tacit information (socialization). Then, process analysts need to validate the hypotheses formulated during the abstraction and formalization of the collected unstructured knowledge (externalization) (*C9*). The knowledge acquisition process is iteratively applied until the formalized version of the knowledge meets the analysis requirements for granularity and modeling perspective, as defined by contextual factors (*C2*, *C8*, *C9*).

However, the currently applied knowledge acquisition process needs support and represents a real bottleneck in the development of business process management. In addition to the acknowledged drawbacks of these approaches, this process is not standardized, and ad-hoc discovery practices make it difficult to create and maintain process documentation. Key challenges include time management for elicitation sessions (*C1*), aligning and standardizing formalized knowledge according to the modeling perspective and purpose (*C2*), support for effective and

[2] Appendix A.3 Problem validation: Interview guide.

adequate preparation before elicitation sessions (*C3*), ensuring quality and validity of elicited information (*C5*), and limitations in semi-formal modeling languages, which render them ineffective for communication with domain experts (i.e., novice modelers) (*C7*). Additionally, the lack of coordination mechanisms among different stakeholder profiles, combined with a top-down strategy, results in insufficient support and engagement from domain experts. Process analysts need to engage in a change management dynamic, relying on collective and individual elicitation sessions (*C4*). These barriers subsequently lead to a lack of maturity and added value in the business process management approach. This may lead to a vicious cycle in which stakeholders increasingly view formal and technical process documentation as unengaging particularly when it is neither used, maintained, nor adapted to their capabilities – resulting in decreased involvement and hindering the successful, organization-wide implementation of a business process management approach (Trkman 2010; Heller and Varney 2013). In order to encourage participation and engagement, it is crucial to support domain experts by guiding them in providing the expected responses, assisting, and enabling them to interact without technical barriers (*C6*, *C9*) or means restrictions (*C7*).

Figure 3 illustrates the knowledge acquisition process performed by business process analysts. In this representation, we identify two key domain expert stakeholder profiles: the *process owners*, who have a high-level end-to-end understanding of process activities and a clear perspective on the coordination and flow between them, and the *field domain experts*, who possess more detailed, low-level process expertise, typically represented at a finer level of granularity. Depending on the organizational context, these individuals and roles might not be distinct. Prior to applying the methodology, a process management project is initiated in collaboration with management, during which the strategic objectives and the project scope are defined. The methodology then begins, guided by these objectives, with the identification of relevant processes and stakeholders, allowing process discovery to start. The business process discovery phase is an iterative approach, incorporating both socialization and formalization.

The interview data analysis identified design requirements within three categories aligned with the knowledge acquisition process. These requirements (see Table 3) are derived from the challenges (*C*) expressed by interviewees.

5 Solution Proposal: Instantiation Design and Implementation[3]

Our proposed solution,[4] *PKAI*, consists of a multi-agent system in which humans (i.e., process analysts and domain experts) interact and collaborate with LLM-based agents to replicate the formalized knowledge acquisition process. The scope, type, and nature of the task significantly influence the performance of LLMs. As these models show limitations in complex reasoning (Fill et al. 2023; Cámara et al. 2023), we opted to partition the process into distinct specific sub-tasks, each managed by tailored agents. Three primary high-level stages (i.e., preparation, socialization, and externalization) are defined to replicate the knowledge acquisition process.

The coordination of the multi-agent systems logic (Fig. 4) highlights the primary interactions between user profiles and (subsets of) agents[5] across the main stages of the knowledge acquisition process. We divide the agent intervention framework into three macro stages (preparation, socialization, externalization), within which we represent the respective roles of process analysts and domain experts. The coordination of the multi-agent system involves providing appropriate knowledge to the relevant agents. The overall task is decomposed into multiple sub-tasks, each carried out by specialized agents interconnected through a shared memory. Each agent's actions sequentially modify the memory, ensuring that all agents operate with the most recent state of the system.

As part of our prototyping approach, we developed a Python-based project hosted on a GitHub code repository.[6] Agent coordination is orchestrated through a custom Python script, which also manages the connection to individual agents via the OpenAI API. The active memory, structured as a Python dictionary (i.e., JSON format), is either fully or partially accessible to the different agents. Table 4 presents the functional utilization of agents by outlining their respective responsibilities.

Preparation replicates the work conducted by a process analyst before engaging with a domain expert. Process analysts use their expertise, available documentation, and implemented support features (see Fig. 5) to tailor a generic interview canvas, adapting it to specific analytical needs. The interview canvas is the foundational framework for structuring the semi-guided interview, consisting of predefined questions. Initially, a set of generic questions is provided, which can be freely modified, expanded, or reordered (*R1–R5*) to define the essential information to elicit and tailor the granularity of the analysis (*R16–R17*).

[3] Demonstration *PKAI*: https://zenodo.org/records/15799395?token=eyJhbGciOiJIUzUxMiJ9.

[4] The technical implementation details are further elaborated in Appendix B and the associated supplementary materials.

[5] The agents shown in Fig. 4. are a subset of agents coordinated to perform different sub-tasks.

[6] GitHub code repository: https://github.com/PeerBlind/PKAI.

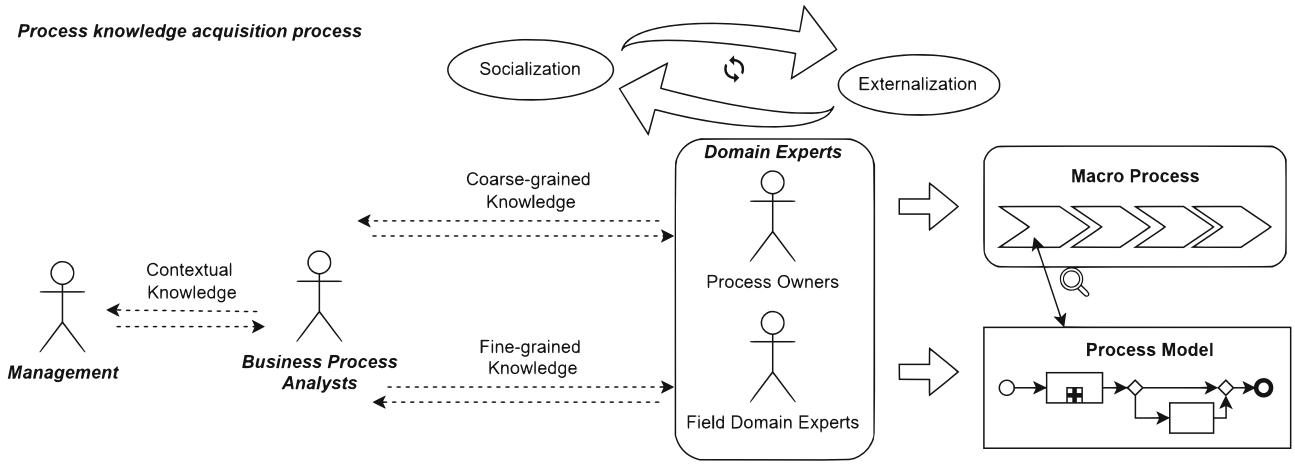


Fig. 3 Knowledge acquisition and formalization process in a business process management context

Table 3 High-level design requirements induced by coding subcategories

Requirements category	Challenges (*C*)	Requirements (*R*)
Preparation	*C1:* Manage elicitation time	*R1:* Process analysts should manage and control elicitation time
	C2: Follow modeling perspective and purpose	*R2:* Elicitation should be adapted to the modeling purpose
		R3: Elicitation should integrate contextual and technical information
		R4: Process analysts should be supported in tailoring the elicitation session
		R5: The elicitation session should guide domain experts in providing the required information
	C3: Provide in-depth context insights	*R6:* Process analysts should be supported in tailoring elicitation sessions based on business and technical information
		R7: Process analysts should be guided and directed with regard to the technical aspects
		R8: The process analyst should be guided in selecting domain experts
Socialization	*C4:* Enable group conversation	*R9:* Elicitation should follow interview or workshop practices
	C5: Monitor information quality	*R10:* The relevance and validity of elicited information should be validated
		R11: Domain experts should visualize and validate the information provided to reduce ambiguities and support coordination
		R12: Elicited information quality should be monitored and ensured
	C6: Provide guidance	*R13:* Domain experts should be guided through the technical aspects of the process
	C7: Simulate human interaction	*R14:* Elicitation mechanisms should replicate a discussion between stakeholders
		R15: To prevent domain experts from being restricted in knowledge transmission, they should be able to express themselves orally
Externalization	*C8:* Adapt granularity	*R16:* The granularity of the process model should be adapted to the modeling purpose
		R17: The granularity of the process documentation should be adapted to the modeling purpose
	C9: Support collaborative modeling	*R18:* Process analysts should monitor, refine, and validate knowledge transmitted or produced by domain experts
		R19: The tasks of domain experts should be comprehensible to audiences of novice modelers

The preparation phase, intrinsically connected to the *research* approach, serves two main objectives: (1) technical preparation, involving domain familiarization, and (2) process-oriented preparation. Based on the uploaded documentation, the business process analyst can request two agents to suggest questions or ask their technical inquiries (*R6–R8*). One focuses on domain comprehension, allowing the analyst to ask targeted questions about

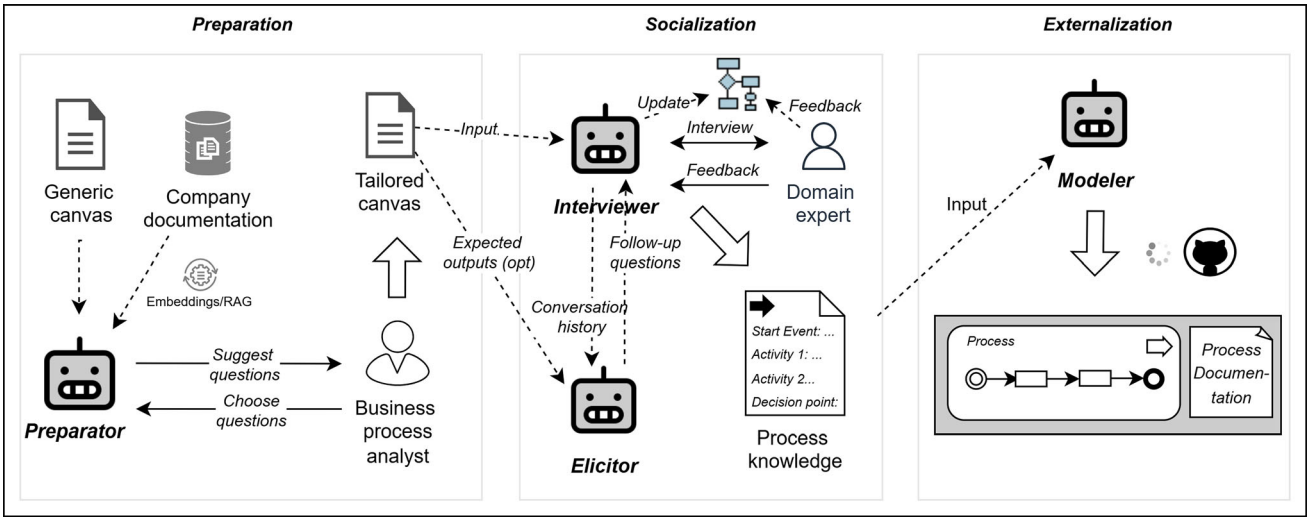


Fig. 4 *PKAI* infrastructure proposal (set of agents are displayed for clarity) – a high-level overview of the sequential application of the knowledge acquisition process

Table 4 Summary of LLM-based agents supporting *PKAI* and their main functions

Agents	Functions	Model	Requirements
Preparator	Generate and answer process technical questions	*gpt-4o*	R6–R8
	Generate and suggest process questions	*gpt-4o*	R6–R7
	Generate and answer process technical questions (RAG)	*LlamaIndex*	R6–R8
	Generate and suggest process questions (RAG)	*LlamaIndex*	R6–R7
Interviewer	Transform text to audio	*whisper-1, tts-1*	(R9)–R14
	Transform speech to text	*gpt-3.5-turbo*	(R9)–R15
	Summarize the process flow	*gpt-4o*	R16–R19
	Mermaid flow chart creation	*gpt-4o*	R19
Elicitor	Checking user willingness or ability to answer	*gpt-4o*	R10–R13
	Checking quality (expected outputs)	*gpt-4o*	R10–R13
	Checking quality	*gpt-4o*	R10–R13
	Generate specific (expected outputs) follow-up questions	*gpt-4o*	R10–R13
	Generate generic follow-up questions	*gpt-4o*	R10–R13
Modeler	Generate process schema (enforced outputs)	*gpt-4o*	R18
	Generate process technical documentation	*gpt-4o*	R16–R19

specific business and contextual aspects. The second agent supports elicitation planning by suggesting specific questions and discussion points. For example, the agent can propose process milestones aligned with theoretical frameworks or management best practices (e.g., ITIL, IT4IT), thereby helping analysts refine and enrich their elicitation strategies. The two tailored agents rely on the indexing of knowledge bases through retrieval-augmented generation, leveraging LlamaIndex and associated libraries for large-scale documentation. In cases involving smaller volumes of knowledge, they instead depend on embedding selected documents identified by analysts as relevant during the preparation phase. These features are illustrated in Fig. 5 with an example of uploading noisy business documentation from a fictional pizzeria. In that scenario, *PKAI* suggests relevant questions to guide the analyst in building the interview canvas (Fig. 6).

From a technical design perspective, document embedding is the default mechanism. However, when documentation volume exceeds a certain threshold (number of tokens), vector database indexing becomes the preferred option to manage token-related costs. More details are available in the supplementary materials.

Socialization consists of an interactive session with the domain experts answering questions from a conversational agent simulating an interview. Domain experts can follow the guidelines of our iteratively enriched (*R13*) implementation home page and then answer orally using vocal mode or traditional chat messages using writing mode (*R14–R15*). The chatbot replicates a semi-guided interview

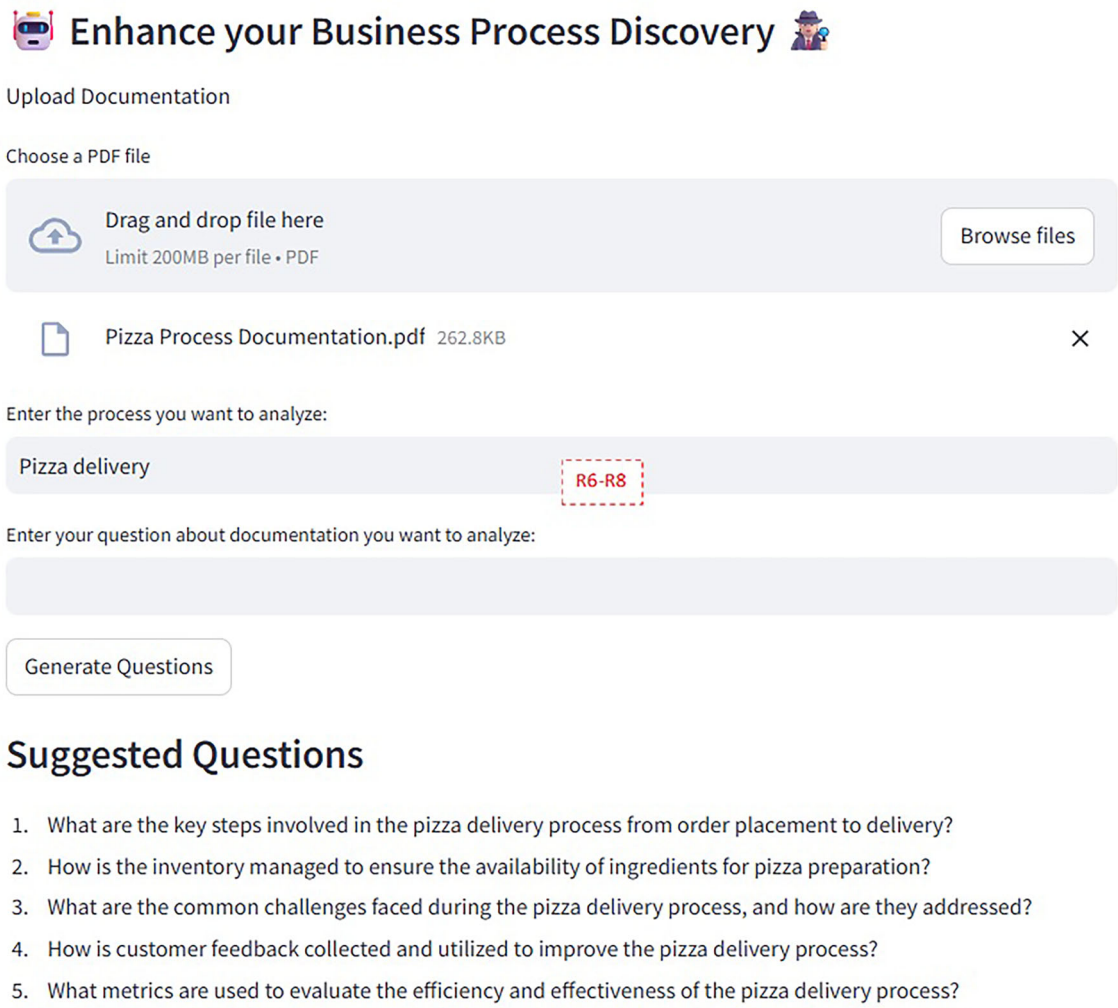


Fig. 5 Enhancing knowledge acquisition preparation phase – Embedded documentation illustration

with two types of questions: defined questions from the tailored interview canvas validated by process analysts during the preparation phase and follow-up questions, generated by our elicitation mechanism detailed below (*R13*). Throughout the interview, led by the Interviewer agent (see Fig. 4), the domain expert can visualize the first semi-formal version of the knowledge transmitted as a flowchart using Mermaid language. Modeling languages such as BPMN introduce complexity and hinder process understanding for domain experts (Ottensooser et al. 2012). During the interviews, they can provide feedback on this model to improve its quality and reduce ambiguity in the information transmitted. Figure 7 illustrates the socialization phase. This model facilitates interaction and enables ambiguities to be clarified or details added (*R10–R11*).

The management of follow-up questions is based on the guidelines provided by Hu et al. (2024). However, we have adapted this framework to align with our specific context and requirements. Multiple agents manage the flow of conversation, determine the need for a follow-up question, and generate these questions. Figure 6 illustrates the design choices underlying this system, emphasizing the sequential intervention of the various agents involved.

First, a dedicated agent assesses whether the respondent is willing or able to answer a particular question and delivers a binary output based on the conversation history. The elicitation mechanism is then designed not to persist with further questions on the same topic. In preliminary tests, users expressed frustration when pressured to address topics, for instance, beyond their expertise.

Next, two layers of verification are implemented (Fig. 6, Appendix B.3. The Appendices are available via http://link.springer.com). These two layers are based on a threshold mechanism driven by LLM outputs assessing the semantic quality of the answer. This threshold is a tunable parameter that can be used by analysts to tailor the desired level of interview granularity: a higher threshold increases the likelihood of follow-up question generation, thereby

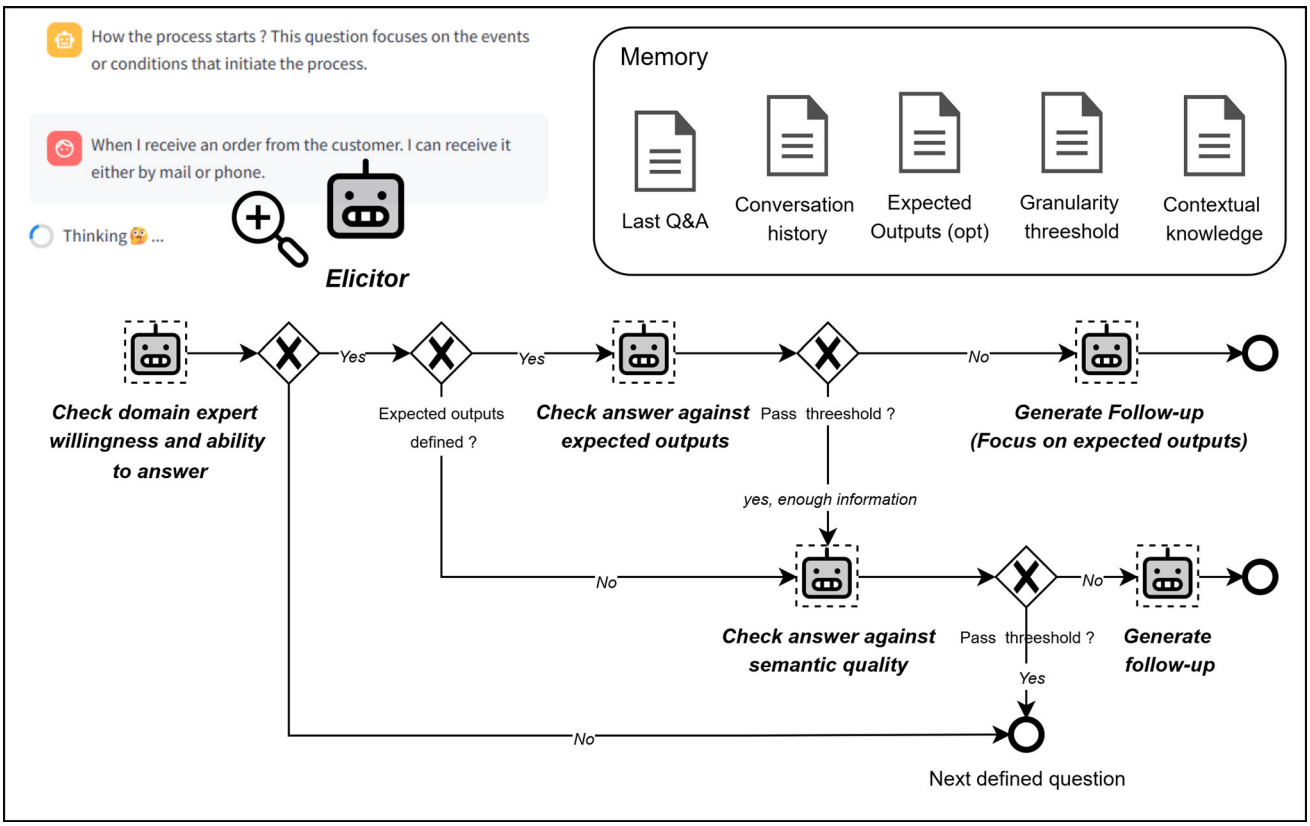


Fig. 6 Conversation monitoring using semi-guided interviews. Inspired by Hu et al. (2024)

producing more in-depth interviews. The first, given priority, activates when the analyst has specified (optional) expected outputs during the preparatory phase. An agent is tasked with evaluating whether the participant's response addresses the elements identified as mandatory by the analyst. If the agent determines that the response lacks sufficient detail regarding the expected output and returns a score below a predefined threshold, another agent will generate a follow-up question focusing on the missing information about the expected outputs. Then, when no expected outputs are specified, a similar evaluation is conducted – this time emphasizing the intrinsic quality and logical consistency of the response. The same threshold mechanism is applied in this third verification step. If the threshold is not met, another agent is responsible for generating a follow-up question, aiming to enrich the user's response. If the score is sufficient, the conversational agent moves on to the next defined question. The interview concludes when defined questions have been answered, no additional follow-up questions are generated, nor do the domain experts provide additional information. The elicitation session is completed once they validate the model representing the high-level process flowchart of information transmitted and have no more details to provide (*R10–R12*, *R19*).

Externalization consists of formalizing the information collected during the interview phase. The first output is a semi-structured process. Documentation is created based on the conversation history, including the modeled process in BPMN format as illustrated in Fig. 8, a high-level textual description of the process flows, an identification of key BPMN elements relevant to the modeling stored in a semi-structured format (*R14–R15*), and the full conversation history with final Mermaid flowchart validated by the domain expert interviewed. This history ensures the process analyst has a traceable record of the information quality and can understand and refine the generated model (*R12*). Beyond model generation, our goal is also to support analysts in interpreting the semantic content of the resulting process models. A raw model is often difficult to understand, especially for process analysts who were not present during the elicitation session. The second output is the BPMN model, provided in an editable.bpmn format (*R16*).

Several LLM-based agents and algorithms are involved in the knowledge formalization process. To generate documentation, conversation histories are transformed by LLM agents using prompt engineering to enforce specific guidelines. To identify BPMN construct components and build the BPMN model, we rely on OpenAI's structured

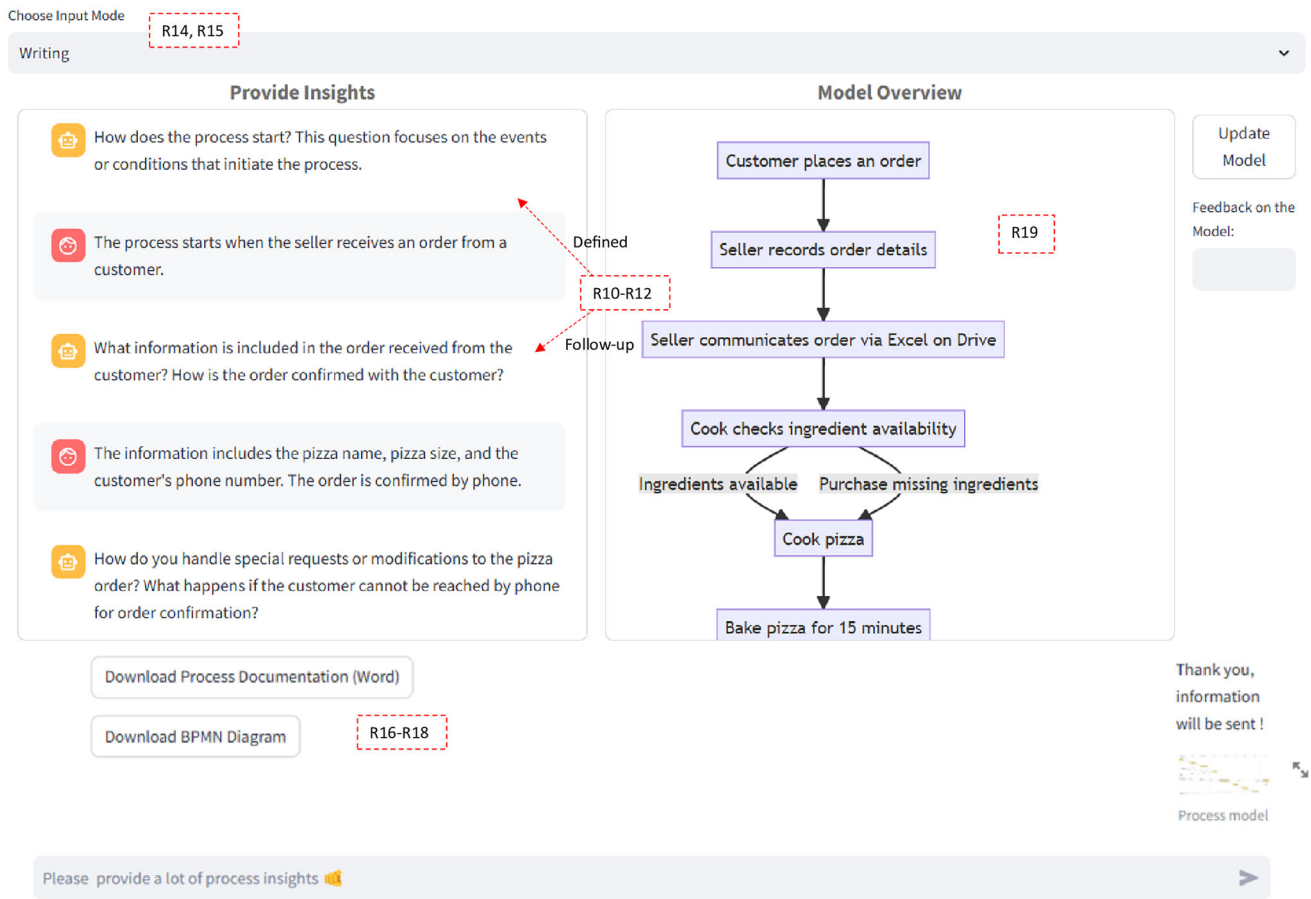


Fig. 7 Process knowledge elicitation with replication of domain expert interviews

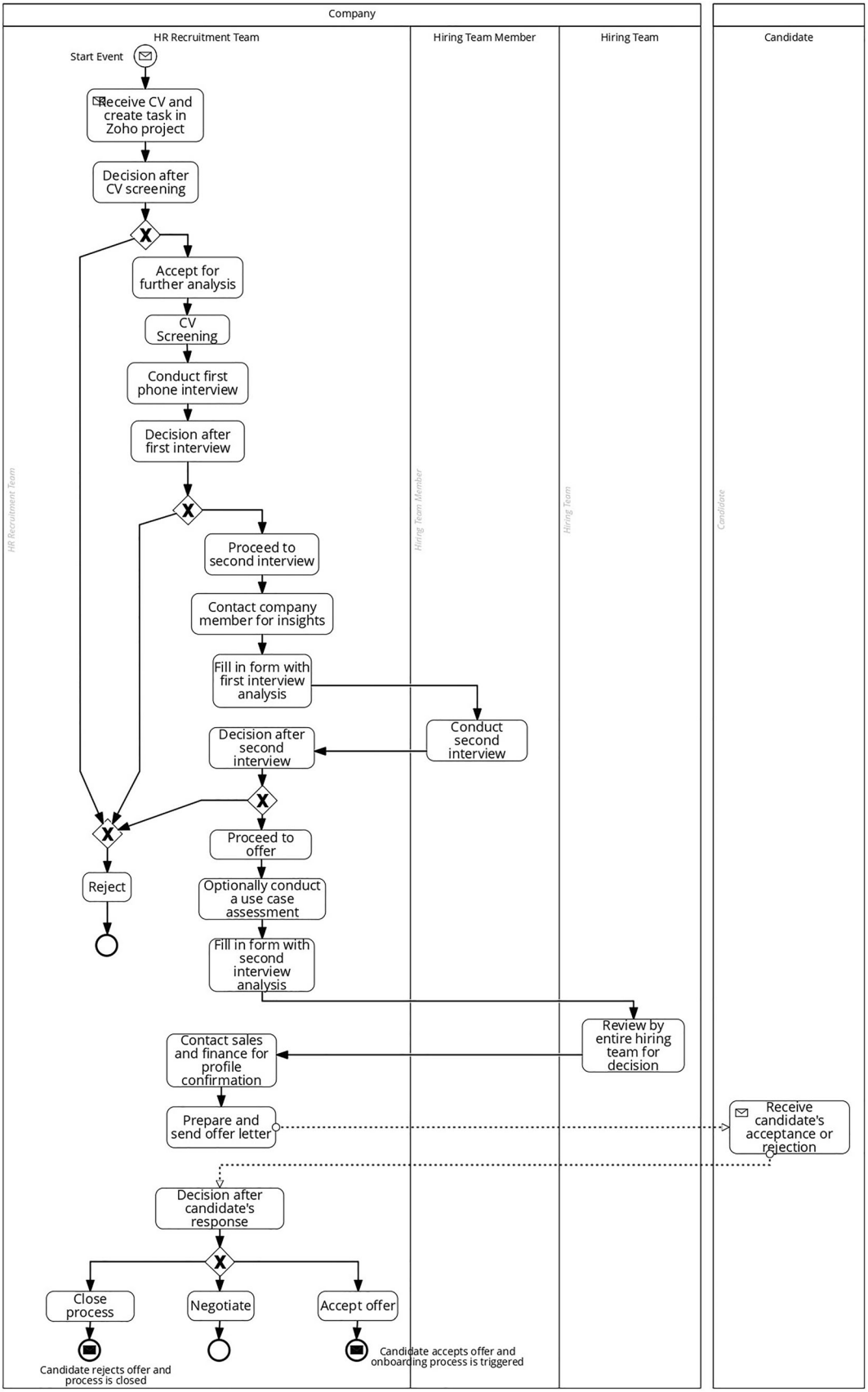
output functionality that follows a predefined JSON schema designed to capture the essential BPMN elements. This approach has proven to be the most efficient in terms of token consumption, while also enabling the transition from unstructured to semi-structured process information (Köpke and Safan 2024).

Then, we rely on Python script algorithms to generate the BPMN model from the collected structured data (i.e., JSON schema). In the first version of our instantiation, the JSON schema is converted into a graph structure using the Python library NetworkX. Then, this graph is transformed into a BPMN 2.0-compliant XML file through an algorithm aligned with the Sketch Miner syntax for BPMN export. Sketch Miner is used specifically to handle mandatory BPMN 2.0 XML tags (e.g., bpmndi) (see Appendix B.4). During the latest iterations of the artifact's design, the transformation process was revised via a Python script that produces XML files compliant with the BPMN 2.0 standard. This algorithm addresses key challenges, including mapping the JSON structure to XML and defining a two-dimensional layout for the elements to ensure that our approach could operate independently of Sketch Miner.[7]

6 Artifact Demonstration and Evaluation

In this section, we first demonstrate the functionalities of *PKAI* to experienced process analysts (Sect. 6.1). Next, we perform a quasi-experimental study based on *PKAI*'s usage to evaluate the quality of the models produced compared to non-supported approaches (Sect. 6.2). Finally, we conduct a real-world case study within an organization to illustrate the potential and the impacts of introducing such an artifact on the knowledge acquisition process (Sect. 6.3). These demonstration and evaluation cycles are part of an iterative process within the DSR methodology employed (Fig. 9). The follow-up interviews and survey took place after *PKAI* instantiation, forming ex-post evaluation (Venable 2006).

[7] Appendix D.2 illustrates the evolution of the models following the latest development iteration of PKAI.


Company
HR Recruitment Team
Hiring Team Member
Hiring Team
Candidate
Start Event
Receive CV and create task in Zoho project
Decision after CV screening
Accept for further analysis
CV Screening
Conduct first phone interview
Decision after first interview
Proceed to second interview
Contact company member for insights
Fill in form with first interview analysis
Conduct second interview
Decision after second interview
Proceed to offer
Reject
Optionally conduct a use case assessment
Fill in form with second interview analysis
Review by entire hiring team for decision
Contact sales and finance for profile confirmation
Prepare and send offer letter
Receive candidate's acceptance or rejection
Decision after candidate's response
Close process
Negotiate
Accept offer
Candidate rejects offer and process is closed
Candidate accepts offer and onboarding process is triggered

◀**Fig. 8** *PKAI* conceptual process model from semi-guided interview with domain experts

6.1 Requirements Validation – Demonstration through Interviews with Business Process Analysts

6.1.1 Data Collection

We presented the artifact with two main objectives. First, to gather feedback on our instantiation choices and verify that the formalized requirements were appropriately addressed. Second, to assess whether such a solution could be integrated into the current practices of process analysts, and get first insights on how this integration might occur. We re-engaged nine of the eleven process analysts previously interviewed and invited them to test our instantiation, explore its functionalities, and react to the generated outputs. Following the demonstration of the functionalities across the different phases – preparation, elicitation, and externalization – we conducted semi-guided interviews (average duration ∼ 35') guided by open-ended questions.[8] After demonstration interviews, participants were provided with a survey based on the "Perceived Usefulness" and "Ease of Use" scales from the TAM model (Davis 1989).

6.1.2 Results

First, the design requirements and mapping with our instantiation were validated. During the demonstration, no difficulties were reported with socialization, and outputs resulting from externalization reached expectations and generated value for process analysts. The survey results indicate a perceived utility score of *5.36* for the solution and an ease-of-use score of *5.44* (on a 1–7 Likert scale). Interviewees acknowledged that these outputs enable the transfer of elicited knowledge. Some asked for tailoring documentation to be returned, which could be developed within our solution without fundamentally altering *PKAI*'s behavior (i.e., "add-on" functionalities). Process analysts emphasized that the initial draft's conceptual model is a promising approach to conceptualizing an understanding of an unknown business process. *PKAI* outputs facilitate comprehension of the overall picture and ensure that business process discovery does not start "from scratch". However, the model must be refined to meet the modeling standards expected by process analysts. In addition, the interviews revealed that the interest in this proposal remains important as long as the tool operates within a clearly defined scope, known in advance, and delivers consistent results.

6.2 Impact on Knowledge Formalization: Quasi-Experimental Evaluation

6.2.1 Data Collection

An in-depth evaluation of the outcomes generated by our artifact was conducted through a quasi-experimental study simulating business process discovery. The study targeted two main objectives: (i) to analyze our proposal's ability to create formalized knowledge based on elicited domain experts' unstructured knowledge, and (ii) to assess the added value of the *PKAI*-generated outputs from the perspective of junior[9] process analysts (n = 26).

The experiment is based on two fictive business cases instantiating processes through discussions and simulated interviews with various organization members. The first is a "holiday booking" process within a travel agency, and the second pertains to a technical repair company. The two business cases[10] are deliberately noisy, containing irrelevant details and mixed levels of granularity to simulate real-world complexity and reflect a complex business environment. Before this experiment ($\boldsymbol{t_0}$), an individual outside our sample, unfamiliar with conceptual modeling techniques, assumed the role of domain expert by answering questions based on the provided business case documentation to generate *PKAI* outputs, which inform the enhanced process analysts (Fig. 10). This experiment's elicitation phase is based on the generic interview canvas, skipping the preparation phase. The experiment was divided into two workshops, each assigned a business case of comparable complexity, with cases and groups subsequently reversed. This approach allows us to consider a larger sample size, thereby enhancing the robustness of the experiment.

In the first phase ($\boldsymbol{t_1}$), Group 1 worked on case A ("holiday booking"). The task was to design a conceptual process model using BPMN, based on the scenarios described in the business case. This phase served as the baseline for our analysis (control group), forming the reference point for the two other experimental conditions involving the use of *PKAI*.

In the second phase ($\boldsymbol{t_2}$), leveraging the participants' familiarity with the business case – akin to the practical knowledge held by domain experts – Group 1 was asked to engage in a *PKAI*-led elicitation interview. Following the

[8] Appendix C.1 Follow-up interview.

[9] Experience ranged from a few months to one year. All participants had a management background and completed training in business process management.

[10] Appendix C.2 Quasi-experimental study.

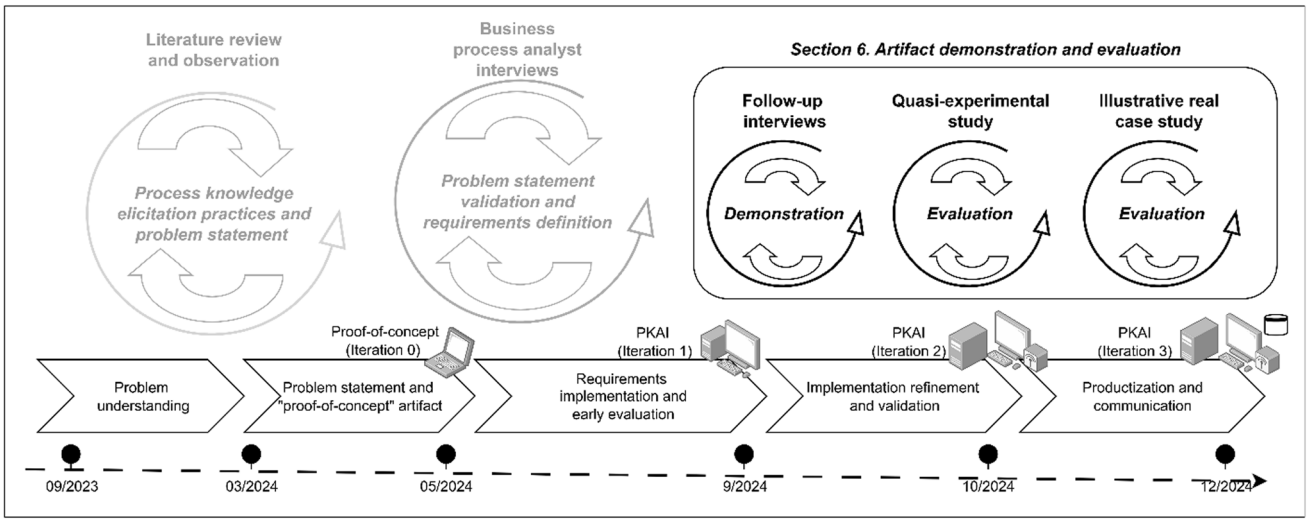


Fig. 9 Solution instantiation development through DSR cycles – focusing on evaluation

guidance of *PKAI*, they needed to provide the process knowledge gained through their experience (simulated by t_1). This interview was structured around a generic question canvas covering the high-level identity of the process. During this stage, participants were no longer responsible for modeling decisions. Instead, their primary task was to convey their process knowledge as unambiguously as possible. This resulted in the automated generation of process documentation. For evaluation purposes, our focus was placed on the process models produced during this phase, while recognizing that a portion of the semantic richness may reside in the other output of *PKAI* – namely, the process documentation accompanying the BPMN model.

In the final phase (t_3), Group 2 received business case A and was given domain-expert-generated documentation using *PKAI* from (t_0). They received the same instructions as the control group but had access to the additional *PKAI* documentation.

For each phase (t_1, t_2, t_3), participants were allocated 25 min, then the same procedure was repeated with a second workshop and business cases reversed ensuring the robustness of comparative analyses. At the end of these two experiments, we conducted two comparisons (Fig. 11): (1) the quality of the models produced by the control group versus those produced by domain experts supported by the artifact (enhanced domain experts), and (2) the quality of the models produced by the control group versus those produced by process analysts supported by pre-generated documentation from our artifact (enhanced process analysts). These scenarios were not arbitrarily chosen but rather derived from our understanding and formalization of the needs collected during the requirement-gathering phase.

After the experiment, we based our evaluation on the 3QM framework to assess the quality of the conceptual models produced. This evaluation and the comparison of aggregated scores were conducted independently for each of the two business cases. This framework was adopted because it enables an objective assessment of model quality, particularly by capturing their semantic value, which is the primary focus of our study. Metrics that fall outside the scope of our artifact – specifically those related to data objects, data object states, and data flows – were excluded from this evaluation. Six batches of process models were considered in our study, along with the application of the 3QM framework within our quality analysis (Fig. 11). A more detailed illustration of the process model quality evaluations is provided in Appendix C. 2. Our evaluation was based on a predefined reference model established prior to the assessment phase. For each business case, the authors defined a list of activities, events, flow conditions, and organizational units. In addition, we identified parallelizable (not belonging to the main flow to consider) flows and macro flows representing the expected high-level sequences of activities. For each category – events, activities, flow conditions, and organizational units – we distinguished between *mandatory* elements (i.e., semantically essential for model relevance) and *optional* elements.

This distinction was necessary given the lack of strict guidelines for process model creation and the potential influence of participants' interpretations on the inclusion of specific elements. This decision impacted our evaluation in the following way: for instance, an optional activity included by a participant was not penalized as "superfluous" (a criterion in the 3QM framework), whereas a missing mandatory activity, from a semantic standpoint,

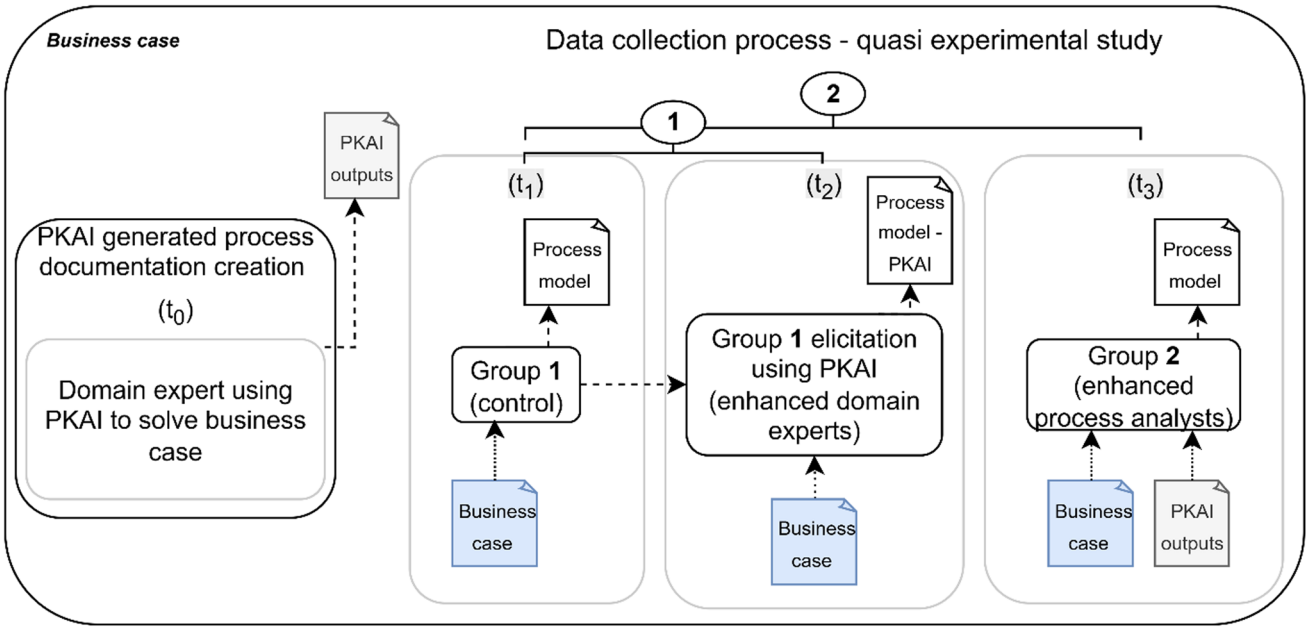


Fig. 10 Quasi-experimental study design, illustration for one business case

was considered a *missing element* and negatively impacted the *completeness* score.

Regarding control flows, we acknowledged that the granularity of the process model could affect evaluation results. To address this, we chose to define macro flows objectively and assessed their presence from a semantic perspective.

These minor adaptations to the 3QM framework allowed for more objective comparisons, aligning our evaluation with the intended purpose of each measure rather than relying solely on raw metrics, which in some cases were less relevant in our context. The transformation of the quality indicators into scores followed the procedures defined by the 3QM framework (Appendix C.2.2). To conduct the two comparisons, we rely on the average of the aggregated model batch scores for each business case independently (Fig. 11).

A final pragmatic consideration: we excluded one participant from Group 1 (n = 10) and two participants from Group 2 (n = 13) due to insufficient engagement with the experiment.

6.2.2 Results

According to the 3QM framework, 1) the quality of the models generated by *PKAI* (enhanced domain experts, i.e., domain expert supported by PKAI) is comparable to those produced by the junior process analysts (control) under the same time constraint (25'). 2) The enhanced business process analysts' model was improved compared to the control group on numerous dimensions. A deeper analysis of these scores (Table 5) reveals that:

Semantics dimension: the *relevance* sub-dimension indicates lower proportion of superfluous elements in the models generated by the tool (higher scores). A similar trend is observed when comparing an analyst working from scratch to one utilizing insights (outputs) from the tool (higher scores). Similarly, the sub-dimension *completeness* is slightly higher for the supported analysts. However, the *correctness* and *flexibility* sub-dimensions show lower scores for models produced by our proposal. In contrast, the correctness scores for enhanced process analysts' models are equal to or higher than those of the control group for these two sub-dimensions.

Pragmatics dimension: the scores for the two sub-dimensions, *unambiguousness* and *understandability*, are equal to or slightly higher in both cases, whether using the tool directly or benefiting from its outputs. Notably, the *understandability* sub-dimension can still be improved algorithmically by the artifact to achieve a level of labeling consistency surpassing that of a human.

Syntactics dimension: models produced by the tool consistently contain fewer syntax errors. In this experiment, however, we observed that the outputs did not necessarily help reduce syntax errors in models refined by process analysts at a later stage, as illustrated by the *text-syntax* sub-dimension.

Given that the primary objective is to replicate knowledge acquisition process and support analysts in creating a final process model, we aim to focus on the semantic contributions provided by its outputs (Table 6).

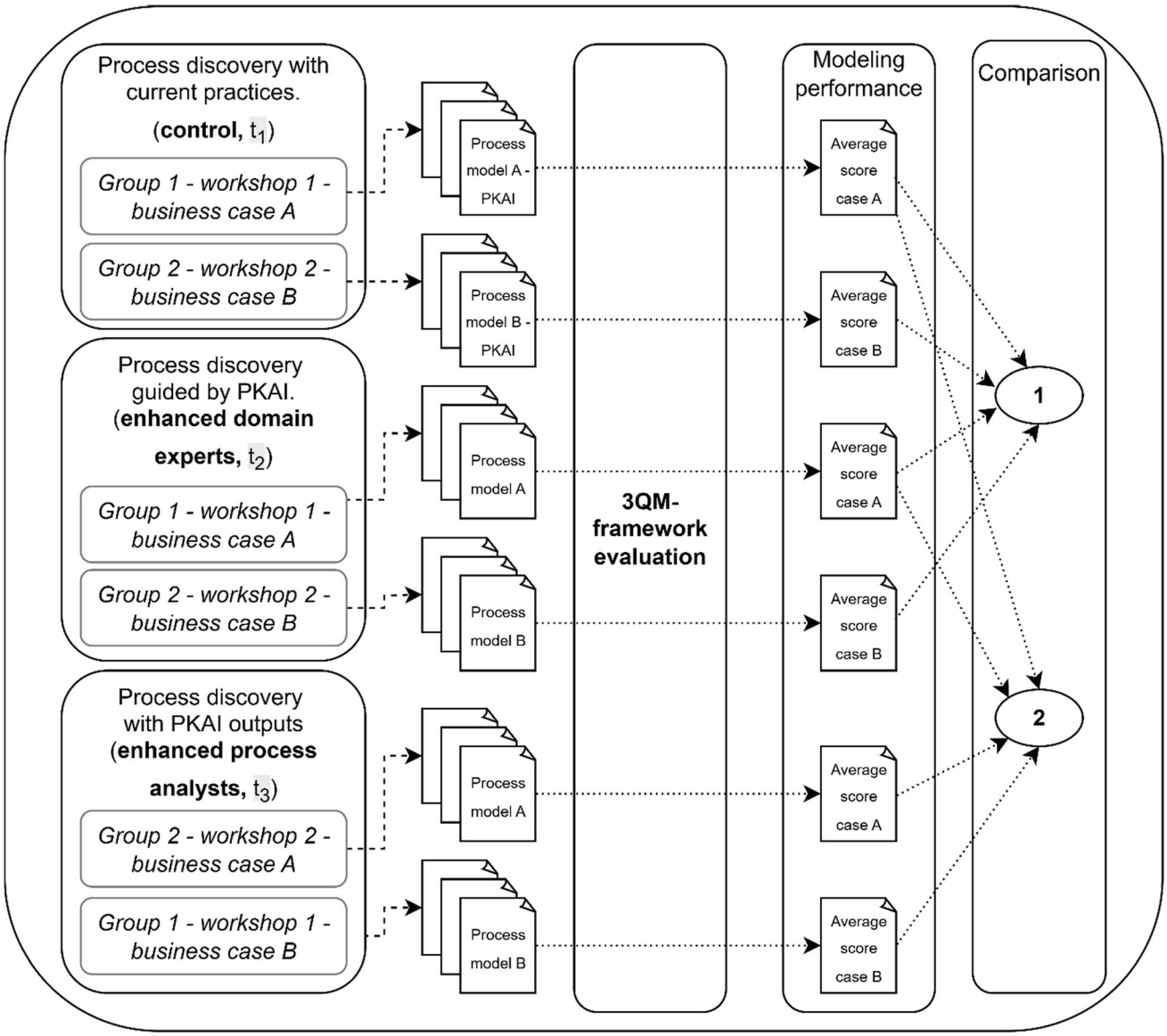


Fig. 11 Quasi-experimental study for *PKAI* evaluation

Analyzing these scores, we can infer that *PKAI*, directly or its outputs, models fewer superfluous elements. It also reduces the number of missing elements, except for flow conditions, when considering the raw model generated by *PKAI*. The proportion of correct elements is higher overall, except for flow conditions and control flows in the raw models produced by the tool. This indicates that the algorithm's understanding of flows is not optimal. However, the outputs, which combine documentation and the model, still allow for equal or superior results in enhanced process analysts' models. Finally, flexibility is reduced in models generated by *PKAI*. Through an analysis of various models, we observed that when two parallel flows, which could be modeled as independent processes, were discussed during the use of the tool, the artifact was not designed to handle the conception of multiple parallel processes. This limitation led to errors in both flow conditions and control flows. Such errors also observed in some control models were, unlike with *PKAI*, not systematic and likely due to the varying experience levels of participants in the experiment.

6.3 Impact on Knowledge Acquisition Process – Illustrative Case Study Evaluation

6.3.1 Data Collection

Finally, *PKAI* was tested in a real-world case study involving actual organizational roles, including an experienced process analyst with four years of expertise in process discovery and three domain experts. The analyst was

Table 5 3QM framework-based scores analysis on conceptual models produced

3QM framework	Sub-dimensions	Enhanced domain experts Vs control (**1**)		Enhanced process analysts Vs control (**2**)	
		Enhanced domain experts (%)	Control (%)	Enhanced process analysts (%)	Control (%)
Semantics	Relevancy	**92**	86	**94**	86
	Completeness	65	64	**69**	64
	Correctness	88	**92**	**96**	92
	Flexibility	69	**95**	90*	**95**
Pragmatics	Unambiguousness	97	97	98*	97
	Understandability	**89**	84	84	84
Syntactics	Text-Syntax	**95**	56	44	**56**
	Sentence-Level syntax	**100**	79	68*	79
	Word – Syntax	**100**	89	**92**	89

*The different scores observed for the two business cases are inversely distributed and cannot be meaningfully interpreted

The bold values indicate higher scores

Table 6 Score analysis of the semantical dimension for conceptual models produced

Semantics	*Metrics*	Enhanced domain experts vs control **(1)**		Enhanced process analysts vs control **(2)**	
		Enhanced domain experts (%)	Control (%)	Enhanced process analysts (%)	Control (%)
Relevancy	Superfluous activity	**91**	80	**89**	80
	Superfluous event	**97**	90	**93**	90
	Superfluous flow condition	**90**	86	**93**	86
	Superfluous Organizational unit	98	98	97	98
	Superfluous control flow	**85**	78	**98**	78
Completeness	Missing Activity	67	65	**77**	65
	Missing event	**61**	57	59	57
	Missing Flow Condition	26	**42**	41	42
	Missing Organizational unit	85	83	**93**	83
	Missing Control flow	**86**	75	74	75
Correctness	Incorrect activity	**94**	91	**96**	91
	Incorrect event	**99**	88	**97**	88
	Incorrect flow condition	85%	**91%**	**94%**	91%
	Incorrect organizational unit	**99%**	93%	**99%**	93%
	Incorrect control flow	61%	**95%**	93%	95%
Flexibility	Constraint	69%	**95%**	90%*	95%

*The different scores observed for the two business cases are inversely distributed and cannot be meaningfully interpreted

The bold values indicate higher scores

tasked with documenting three human resource management processes within a small consulting firm – namely hiring, onboarding, and payroll management – and therefore conducting three elicitation sessions with different domain experts identified as relevant to each process. Following these sessions, one of the authors attended in the three workshops that led to the validation of the final version of the documentation. This experiment aimed to illustrate the impact on the knowledge acquisition process, provide insights into usage practices, and validate our proposed solution within a real, complex business environment involving actual processes. Furthermore, to complete the quasi-experimental evaluation in a real setting, we evaluated final process model produced by PKAI using the semantic dimension of the 3QM framework. We followed observation guidelines from van Maanen (2011) during this experiment without impacting the modeling choice and *PKAI* use.

6.3.2 Results

During the preparation, the business process analysts relied on supporting functionalities of PKAI to define the discussion points. For instance, specific questions identified include: "How is the employment contract signed, stored, and managed?" (hiring) or "What systems or tools are used to execute payment transfers (e.g., ERP, bank payment system)?" (payroll management).[11] Then, semi-guided elicitation sessions with PKAI were entirely delegated to domain experts through a web application link. The first insights showed that the interviews were more structured and standardized. The answers are straight to the point and concise, and the granularity was aligned with the question. They no longer feel compelled to delve into specific problems, but instead focus on an abstract and objective representation of their activities. We also observed that some domain experts took advantage of the asynchronous nature of the discussion to independently verify more specific points (i.e., checking documentation). *PKAI* outputs were then returned to the analyst, who started refining the conceptual models.

Conceptual models generated by LLMs (Fig. 8, illustrated) can be considered as initial formalized knowledge representations. While specific observed patterns require improvement and the generated models rely on a simplified version of the BPMN language, the added value lies in transmitted knowledge creation. This significantly enhances the early exploratory phase by empowering domain experts to independently generate preliminary formalized knowledge sketches in a language they do not fully master, thereby facilitating the initial abstraction of tacit and unstructured knowledge.

Process analysts have identified the primary utility of *PKAI* in the exploratory phases of knowledge acquisition but can use *PKAI* iteratively to analyze processes while reviewing and refining the formalized knowledge. Figure 12 shows the tasks supported by the iterative application of the knowledge acquisition process.

We analyzed the evolution of the semantic richness of process models to evaluate *PKAI* outputs' added value. This analysis demonstrates (see Table 7) that the deliverables produced by *PKAI* following the simulated elicitation enable process analysts to obtain a significant understanding of the targeted processes and generate advanced-stage formal documentation.

Our observations and the various scores for the semantic dimension also highlight several implementation challenges (IC). First, some part of the tacit knowledge can be distributed across several stakeholders. Further in-depth investigation is warranted regarding the integration of our artifact or a comparable proposal into highly granular processes characterized by the distribution of knowledge across multiple domain experts (IC1: Multiplicity of stakeholders).

Another challenge in data collection concerns the regulation of the granularity of the information gathered. The individuals interviewed who were using PKAI provided a level and depth of information that enabled process analysts to critically examine certain validation points during a dedicated workshop. However, in certain cases, where the processes could be more granular, it may be necessary to fully elaborate on sub-processes that were not initially described. These results highlight the artifact's limitation in probing deeper into details when necessary and in navigating between different levels of granularity, as a human analyst would typically do during a traditional interview (IC2: granularity management). Although questions were formulated to influence the desired level of detail, granularity of the responses often varied based on the profile of the individual being interviewed and their subjective perception of the relevance of detailing specific steps (IC2). Additionally, the models generated by *PKAI* are high-level and lack detail. They fall short of the semantic richness needed to achieve the completeness required by analysts (IC3: completeness management). We observe that the instances of missing information are due to a lack of detail provided by the domain expert. One possible reason is that domain experts may face difficulties in interpreting the questions posed. This encompassed both the technical aspects and the overall scope of the inquiry (IC4: domain experts' empowerment). Another possible explanation could be that the relevant knowledge resides with other domain experts who were not consulted by *PKAI* (IC1). This lack of information skews the relevance and correctness scores, as these metrics are based on a less-detailed (complete) model.

By analyzing the models refined by the analyst, we observe that the limitations are mitigated when the PKAI model is combined with the accompanying documentation and the analyst's expertise. This integration enables the construction of semantically rich conceptual models, even with respect to their completeness. Based on the outputs (see Table 7), the business process analysts can refine the initial model and achieve a highly advanced version. Certain aspects were subject to validation, and drawing on their experience, the analyst identified elements overlooked during the elicitation session and prepared for the validation session with these points in mind (Fig. 13, validation points in red). Overall, refined model scores demonstrate that *PKAI* does not introduce ambiguity when the analyst knows how to position *PKAI*'s contribution to their work.

[11] Questions and answers can be found in Appendix C.3 and associated supplementary materials.

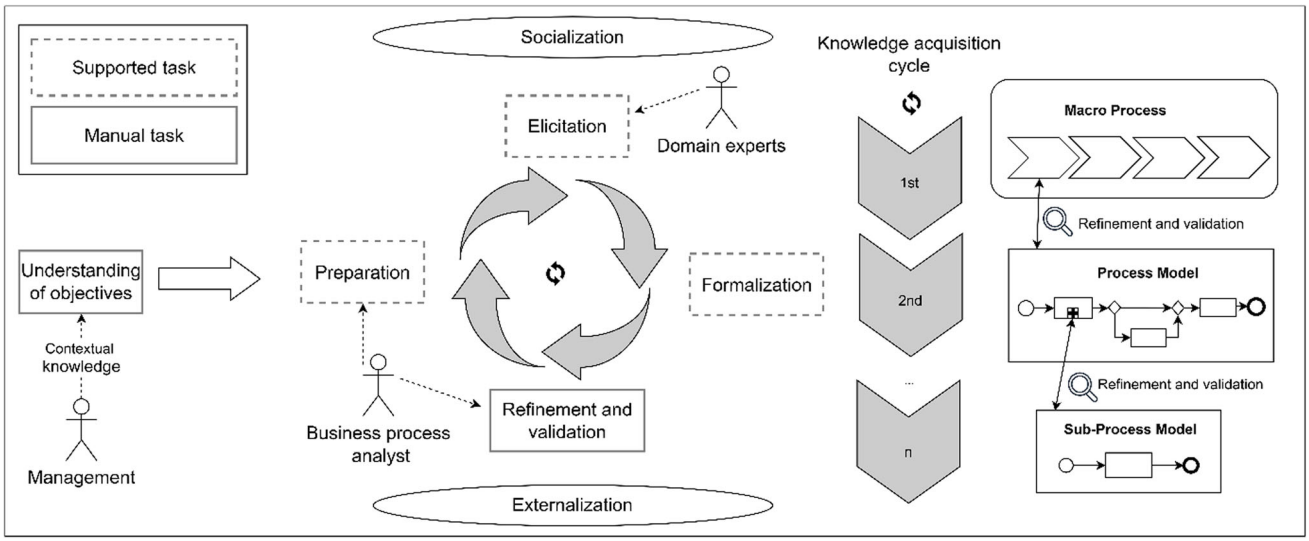


Fig. 12 Integrating *PKAI* to support business process discovery

Table 7 Conceptual models' evolution through the knowledge acquisition process

Semantics	*PKAI* drafts (%)	Refined models (%)
Relevance	**89**	85
Completeness	44	**91**
Correctness	**96**	85
Flexibility	42	**67**

The bold values indicate higher scores

7 Discussion

7.1 Theoretical Contributions

(RQ1) What are the process knowledge acquisition challenges that LLMs can support? Applying the DSR methodology, we redefined process discovery in terms of theories of knowledge creation and acquisition. We formalized process analysts' challenges into design requirements, which, to our knowledge, have not been previously used to support the end-to-end knowledge acquisition process in business process management. While our evaluation focuses on this approach, the design requirements also contribute to the abstraction of generalizable conceptual knowledge. Our proposal is an instantiation (Winter 2008) of a broader issue concerning knowledge acquisition within organizations, and these design requirements could be applied across various domains (e.g., knowledge acquisition beyond business process management).

(RQ2) How to design LLMs-supported process knowledge acquisition? The integration and orchestration of the various stages of the knowledge acquisition process through a multi-agent system constitutes a novel design contribution. It establishes a flexible foundational architecture for future design iterations and empirical investigations into the use of generative AI to support knowledge acquisition. The study identified two key factors that were instrumental in enabling this support:

The iterative refinement during socialization enhances the semantic and pragmatic quality of elicited information: Inspired by Hu et al. (2024), our framework generates follow-up questions to improve the quality of elicited information but ensures a clear direction for elicitation through defined questions. Existing proposals rely on stimulating elicitation without guiding discussions on technical aspects (e.g.,Brown et al. 2011, 2014; Harman

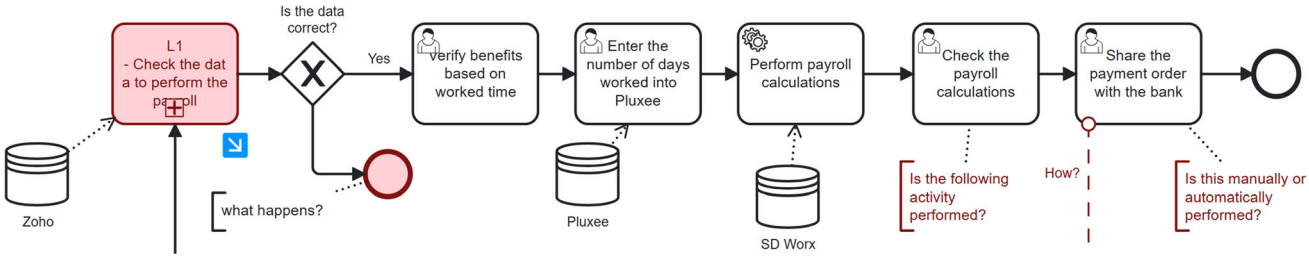


Fig. 13 Excerpt from a refined model introducing validation points

et al. 2015, 2016; Simões et al. 2016, 2018). The semi-structured interview approach bridges the gap between tacit domain expertise elicitation and the pragmatic needs of process analysts, enabling the formalization of actionable process knowledge. Moreover, while literature proposals overlook iterative aspects of elicitation, our solution separates the sequential flows of elicitation from the externalization, enabling stakeholders to discuss some previously missing points without compromising the quality of the formalized information flow. Following guidelines from Kathleen et al. (2014), our approach associates textual description with a conceptual model adapted to domain experts' capabilities, enabling them to refine the knowledge they transmit. Consistent with prior literature (Mendling et al. 2010), information abstraction allows stakeholders to participate effectively, thereby improving validation and reducing ambiguities. As *PKAI* empowers autonomous domain experts to share knowledge and even initiate its formalization, it challenges the traditional scope of their involvement during process discovery.

LLM-generated BPMN conceptual models and documentation eliminate the need to start from scratch for externalization: LLMs offer significant potential for supporting process knowledge formalization. Although the drafts generated may lack intricate details, they provide well-structured high-level outlines. This enables a reexamination of process discovery approaches by reconsidering certain supported tasks, thereby broadening the scope to include new conceptual frameworks. Our proposal, grounded in a deeper understanding of stakeholder-specific needs and differentiated by role profiles, stands apart from existing literature in several key aspects. First, we state that domain experts require guidance throughout the elicitation process; mere stimulation is insufficient to ensure the pragmatic relevance of the information conveyed (Simões et al. 2016, 2018; Antunes et al. 2020; Thuan et al. 2022). Second, we observe that experienced analysts primarily require support in creating an initial structured version of knowledge, which is not solely based on models, especially when they are incomplete. This step must actively involve the domain experts. Through our empirical studies, we recognize that the added value of using LLMs to support process discovery does not lie in assisting with the modeling of knowledge that has already been elicited and semi-structured (i.e., text to model proposals;Köpke and Safan 2024; Kourani et al. 2024), especially given the current state-of-the-art capabilities in generating conceptual models. For instance, while the functionalities described by Ziche and Apruzzese (2024) are designed to facilitate modeling tasks, an experienced process analyst would derive limited benefit from such support, while a non-guided domain expert cannot meet the technical requirements of conceptual modeling.

(RQ3) How does introducing LLMs impact the knowledge acquisition process? Our paper is the first to offer empirical knowledge about LLM-supported business process discovery and comparison with current manual approaches. This research demonstrates that an LLM-based multi-agent system can support the knowledge acquisition process. Evaluations validate our design requirements, demonstrate the formalization of semantically richer process models within comparable timeframes for junior process analysts, and illustrate the added value for experienced process analysts in a concrete business environment.

Nonetheless, from a methodological standpoint, this raises further questions regarding how the artifact should be introduced or how the methodological guidelines should be adapted to optimize its effectiveness (related to the implementation challenges identified in Sect. 6.3). Addressing these challenges constitutes a key direction for future work and will benefit from additional empirical investigation. As discussed in the literature and observed in our empirical studies, *PKAI* was deemed unsuitable for tasks requiring strategic reasoning, such as determining modeling perspectives or granularity levels (Fill et al. 2023; Cámara et al. 2023). This reinforces the notion of a technology driven approach complemented by the critical thinking of a business process analyst.

The documentation generated by *PKAI* provides an overview for identifying and addressing the semantic elements in the model that require further refinement. In practice, this iterative approach to validating the model and documentation is not unique to *PKAI* use, but rather, it aligns with the knowledge acquisition process. Process documentation traditionally follows iterative steps of exploration and knowledge elicitation (socialization), as well as the formalization and validation of that formalized knowledge (externalization), until the quality and granularity of the documented models meet the analytical needs dictated by contextual factors (Fig. 12).

7.2 Threats to Validity and Future Research

This section outlines potential threats to the validity of the knowledge claims underlying our design science research, organized according to the design science validity framework (Larsen et al. 2025).

7.2.1 Criterion Validity Threats

Threat to criterion efficacy validity: our proposal does not incorporate state-of-the-art elicitation capabilities. Specifically, it relies on knowledge explicitly communicated by domain experts and does not capture "invisible actions" that automated process discovery techniques such as

process mining could detect (van der Aalst 2010, 2012). To stimulate elicitation, no additional techniques (e.g., 3D storyboards, process story scenes) are employed; information collection is managed in real time via follow-up generation. During socialization, stakeholders may omit elements perceived as intuitively obvious, yet inaccessible to AI agents, thus impeding completeness. For example, handling implicit process flows remains a key challenge.

We currently support only a reduced subset of BPMN elements, which simplifies model generation but limits expressiveness compared to the full BPMN 2.0 specification. Certain modeling limitations persist – for instance, the distinction between pools and swim-lanes. Our proposal lacks a formalized rationale for distinguishing independent actors and our instantiation provides a hard-coded limitation of three swim-lanes per pool (see Fig. 8). Furthermore, several enhancements remain underdeveloped: support for group-based elicitation (e.g., workshop mode or multiple stakeholders involved) and granularity control is currently suboptimal. These limitations collectively prevent the artifact from performing as well as an experienced analyst. However, future research could integrate these improved elicitation and modeling techniques through our modular multi-agent architecture. Our solution can be improved by drawing on state-of-the-art literature proposals and empirical studies. For instance, we could assess the quality of the elicited information obtained through our supported semi-guided interviews and compare it with various elicitation techniques proposed.

Threat to criterion characteristic validity: our study is grounded in a detailed understanding of an identified need and problem. Despite multiple data collection efforts and iterative evaluation cycles, some degree of subjective interpretation remains when translating observed needs into formalized requirements. The demonstration cycle relied on feedback obtained through the TAM scale and follow-up interviews, but this evaluation was based on the isolated use of the artifact in a fictitious scenario. Thus, the assessment of functionalities may not fully reflect real-world application. In terms of externalization, we focus on process-oriented knowledge formalization, aligned with conventional documentation standards and modelling practices, based on the 3QM framework from Overhage et al. (2012). Although the artifact is currently designed for this type of representation, its architecture could support different abstractions. Above all, generic design principles (Gregor et al. 2020) are a logical next step in the research of LLM-supported knowledge acquisition. Specifically, this artifact paves the way for the application of reflective design principles, as described by Möller et al. (2020).

7.2.2 Causal Validity Threats

The demonstration phase represents the only instance where individual features were evaluated. As such, we need further empirical evidence to isolate the causal effects of specific features, for instance those related to preparation support or the interaction between flowcharts and domain experts during socialization. Some observed outcomes are broadly attributed to the artifact without rigorous empirical grounding. For example, improved structuring of knowledge is credited to *PKAI* as a whole, although this may result from specific mechanisms such as follow-up generation or the behavior of the conversational agent, elements that could operate similarly in other contexts independent of *PKAI*. Moreover, certain capabilities, such as managing ambiguous, incomplete, overly granular, or insufficiently granular knowledge, were not assessed and thus cannot be generalized from our current evaluations. This calls for extended empirical investigation to assess the generalizability and robustness of the findings.

7.2.3 Context Validity Threats

Threat to ecological validity: our evaluations do not explicitly address how ambiguity in expert-provided information affects the artifact's capacity to generate semantically coherent outputs. This warrants further investigation as the complexity scalability of the LLMs used remains unassessed (Fill et al. 2023; Cámara et al. 2023). The quasi-experimental study was constrained by a short timeframe (3 × 25 min), limiting analysts' ability to fully leverage *PKAI* outputs. The simplified processes used do not reflect the complexity of real-world operations. These constraints influenced the quality of the models produced and may have biased evaluation outcomes. For example, junior analysts may prioritize semantic richness over adherence to syntactic standards and modelling best practices (Mendling et al. 2010). Additionally, the participants lacked professional modelling experience but demonstrated strong abstraction capabilities, which may have influenced their performance as domain experts despite natural language-based conversation.

In the illustrative real-world case study, participants held managerial roles with an overview of end-to-end processes. This allowed coherent modelling from single-person interviews, a situation unrepresentative of highly granular processes in which tacit knowledge is distributed across multiple stakeholders. Additionally, the study focused on standard HR support processes, which are relatively common across organizations and familiar to analysts, thereby limiting the generalizability of the findings compared to more operational processes that are specific to the analyzed company. It would be valuable to investigate

whether our proposal is complexity-scalable – both in the complexity of processes and their representation (e.g., highly granular processes). These limitations of current investigations are shared by Cámara et al. (2023) and Teubner et al. (2023). Further evaluations are needed to assess *PKAI* effectiveness with several analysts' and domain experts' profiles. In future research, we will deploy *PKAI* in organizational settings to evaluate its scalability and performance across user profiles and process complexity. This would help to actualize the potential of LLMs for each phase of the knowledge acquisition process.

Threat to external validity: the elicitation of needs, definition of requirements, and demonstration-based evaluation were all designed around practitioner needs. However, these practitioners primarily operate in northwestern Europe and in companies with subsidiaries in Belgium. The generalizability of our findings across regions and organizational contexts remains uncertain. We hypothesize that differences in process maturity and culture could significantly affect both the validity and transferability of the results. Finally, the artifact's agentic components rely on proprietary OpenAI models. While effective, this introduces concerns regarding the protection of sensitive knowledge and data. Moreover, behavioral variation across LLM providers remains unexplored. Future work should incorporate open-source models (E.g., Mistral) to evaluate alternative configurations.

8 Conclusion

Current manual business process discovery approaches face several limitations. This paper provides the first LLM-based artifact instantiation spanning the whole knowledge acquisition process, empirically highlighting its benefits and limitations around the design contributions structure (Venable 2006).

First, we define a formalized understanding of a generic problem related to knowledge acquisition in business process management. The formalization of the knowledge acquisition process allows us to define a set of design requirements related to the diagnosed problems. Second, we design and implement an instantiation of an artifact aligned with our design requirements by combining various constructs. This multi-agent infrastructure replicates the knowledge acquisition process enhanced by LLMs. This solution is instantiated within the context of business process management and enables the socialization and externalization of process knowledge. Finally, this proposal is evaluated through three cycles, enabling us to issue knowledge claims about the use and impacts of LLM-based process discovery. Follow-up interviews confirm the effective formalization of the challenges, and that our artifact fully addresses the design requirements. A quasi-experimental study shows that the models produced through *PKAI* are more complete and relevant than those created by junior process analysts. The knowledge conveyed also enables the creation of semantically richer models. An illustrative case study demonstrates that process analysts can agilely integrate *PKAI* into their practices, allowing domain experts to perform specific exploratory to pre-formalization tasks.

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